Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 6 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 6 : SLO â€“ 1 | Matrix Multiplication with DMA â€” Allocation

I. Fill in the Missing Code

1. Declarations: int **mat1, **mat2, **res; int i, j, k; int r1, c1, r2, c2;
2. Looping variables: int i, j, k;
3. Dimensions: printf("Enter rows & cols of mat1 and mat2"); scanf("%d %d %d %d", &r1, &c1, &r2, &c2);
4. Validity check: if (c1 != r2) { printf("Multiplication not possible\n"); return 0; }
5. mat1: mat1 = (int**) malloc(r1 * sizeof(int*)); for (i = 0; i < r1; i++) mat1[i] = (int*) malloc(c1 * sizeof(int));
6. mat2: mat2 = (int**) malloc(r2 * sizeof(int*)); for (i = 0; i < r2; i++) mat2[i] = (int*) malloc(c2 * sizeof(int));
7. res: res = (int**) malloc(r1 * sizeof(int*)); for (i = 0; i < r1; i++) res[i] = (int*) calloc(c2, sizeof(int));

II. Notes

• malloc for input matrices (values will be read in), calloc for the result so accumulators start at 0.
• Always check c1 == r2 before allocating the result matrix.